Joseph Lee

Data Scientist | Applied Machine Learning and Evaluation | Large-Scale Data Modelling
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SUMMARY

Data scientist with a background in applied machine learning and data analytics. Experience benchmarking and stress-testing predictive models against ground-truth data and diagnosing systematic failure modes. Experienced in building and evaluating predictive models in Python and benchmarking performance under real-world constraints.
Strong analytical and communication skills to translate complex data into actionable insights for decision-making.

SKILLS

Machine Learning & Modelling: Scikit-learn, TensorFlow, PyTorch, Bayesian modelling, Gaussian Processes
Programming: Python (NumPy, pandas, matplotlib, seaborn), R, SQL, Tableau
Data management & Infrastructure: HPC, Docker, Unix, WSL, Git

EXPERIENCE

Junior Data Scientist 09/2026 – PRESENT
Akeso
- Contributing as part of a client-facing data science team working across healthcare and life sciences, including maintenance and development of an LLM categorisation tool that maps unstructured hospital equipment data to standardised hierarchies, reducing classification effort and enabling spend analysis for healthcare clients

Postgraduate Researcher 05/2025 – 09/2025
Lovell Lab, University of Manchester
- Identified a consistent false positive in generative binder predictions, where implausible binders were predicted as viable in low-information conditions, motivating incorporation of sterically and chemically aware constraints to improve model specificity and performance in real-world drug discovery

Postgraduate Researcher 01/2025 – 05/2025
Rattray Group, University of Manchester
- Optimised a Bayesian Machine Learning approach for time-series analysis for ~20,000 genes, identifying ~4,450 previously undetected expression patterns from traditional models
- Communicated model results and their statistical assumptions to non-technical stakeholders, informing downstream decision-making experiments

R&D Analyst (part-time) 06/2025 – PRESENT
Patent Outsourcing Ltd
- Delivered written patent and literature analyses across UK, EU, and US jurisdictions for biotech clients, translating complex scientific data into actionable insights for non-specialists

Associate Scientist (Protein Engineering) 01/2021 – 08/2023
Immunocore
- Statistical analysis and evaluation of internal drug databases (3000+ molecules) to benchmark high-performing pharmaceutical candidates and support selection decisions
- Designed and executed high-throughput experiments (1200+ candidates) to optimise specificity across drug discovery programmes

PERSONAL PROJECTS

Corridor of uncertainty | Python, Git, OpenAI API
- Schema-driven LLM evaluation model with structured JSON outputs for contextual reasoning in specialised jargon (vs generic benchmarks like MMLU) with deterministic per-field scoring across multiple models, uncovering strong factual recall but poor inference of implied outcomes

Protein Language Model | Python, Git, ESM2, MMseqs2, TensorFlow, JAX/FLAX
- Built a protein function prediction model using ESM2 embeddings, TensorFlow and JAX/FLAX architecture, improving recall by 76% and AUPRC by 56% compared to baseline through sequence augmentation, MMseqs2 clustering and stratified group k-fold cross-validation

EDUCATION

University of Manchester 09/2024 – 09/2025
MSc Bioinformatics & Systems Biology **Grade: Distinction (79%)**
- Modules: Programming (84%), Computational Biology (81%), Research Project 2 (80%), Research Project 1 (78%), Bioinformatics (68%), Statistics (Pass/Fail) (Pass)

University of Bath 09/2017 – 06/2021
BSc (Hons) Molecular and Cellular Biology **Grade: 2:1 (67%)**
- Dissertation (71%): Statistical analysis of public SARS-CoV-2 datasets to assess intranasal IgA vaccination strategies
- Activities: Italian (CEFR level A2), University of Bath Cricket Club: Secretary, University of Bath Rugby Club